

习题 2

1. 判定下列方程的类型:

(1) $4u_{xx} - 7u_{xy} + 3u_{yy} = 0$;

(2) $a^2u_{xx} + 2au_{xy} + u_{yy} = 0$, a 为常数;

③ $x^2u_{xx} - y^2u_{yy} = 0$;

④ $x^2u_{xx} + (x+y)^2u_{yy} = 0$.

2. 化下列方程为标准型:

① $u_{xx} + 4u_{xy} + 5u_{yy} + u_x + 2u_y = 0$;

② $x^2u_{xx} + 2xyu_{xy} + y^2u_{yy} = 0$;

(3) $u_{xx} - 2\cos xu_{xy} - (3 + \sin^2 x)u_{yy} - yu_y = 0$;

(4) $(1+x^2)u_{xx} + (1+y^2)u_{yy} + xu_x + yu_y = 0$.

3. 求下列定解问题的解:

①
$$\begin{cases} u_{xx} + 2\cos xu_{xy} - \sin^2 xu_{yy} - \sin xu_y = 0, & (x, y) \in \mathbb{R}^2, \\ u(x, \sin x) = x + \cos x, \quad u_y(x, \sin x) = \sin x, & x \in \mathbb{R}^1. \end{cases}$$

(2)
$$\begin{cases} u_{xx} - 2\sin xu_{xy} - (3 + \cos^2 x)u_{yy} + u_x + (2 - \sin x - \cos x)u_y = 0, & (x, y) \in \mathbb{R}^2, \\ u(x, \cos x) = 0, \quad u_y(x, \cos x) = e^{-x/2} \cos x, & x \in \mathbb{R}^1. \end{cases}$$

4. 证明: 两个自变量的二阶常系数双曲型方程 (2.1.1) 一定可以经过自变量及未知函数的非奇异变换 $u = Ve^{\lambda\xi + \mu\eta}$ 将它化成

$$V_{\xi\xi} - V_{\eta\eta} + CV + F(\xi, \eta) = 0$$

的形式, 其中 C 是常数, F 是已知函数.

5. 适当选取参数 λ 和 μ , 利用变换 $v(x, y) = e^{\lambda x + \mu y}u(x, y)$, 化简下列方程:

(1) $u_{xx} + u_{yy} + \alpha u_x + \beta u_y + \gamma u = 0$;

(2) $u_{xx} = a^{-2}u_y + \beta u_x + \alpha u$;

(3) $u_{xx} - a^{-2}u_{yy} = \alpha u_x + \beta u_y + \gamma u$;

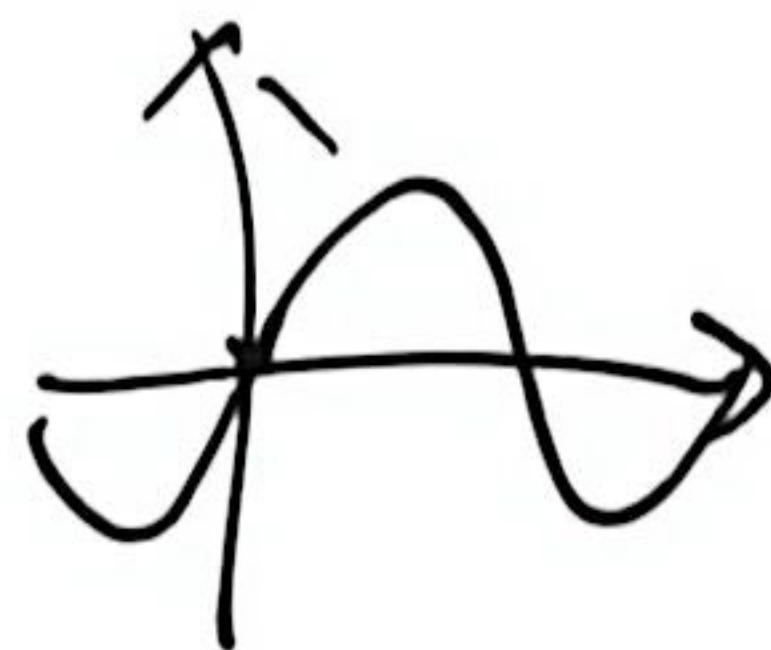
(4) $u_{xy} = \alpha u_x + \beta u_y$.

6. 求下列方程的通解:

$$u_{xy} = \frac{1}{x-y}(u_x - u_y).$$

7. 设方程 (2.1.29) 中的系数 A, B, C 满足 $B^2 - 4AC = 0, A \neq 0$. 证明方程 (2.1.29) 的通解具有如下形式:

$$u(x, y) = f(mx + y) + xg(mx + y),$$



其中 f, g 是任意两个二阶可微函数和 $m = -\frac{B}{2A}$.

8. 求下列方程的标准型:

$$(1) u_{xy} - u_{xz} + u_x + u_y - u_z = 0;$$

$$(2) u_{xx} + 2u_{xy} + 2u_{yy} + 2u_{yz} + 2u_{yt} + 2u_{xt} + 2u_{tt} = 0;$$

$$(3) \frac{\partial^2 u}{\partial x_1^2} + 2 \sum_{k=2}^n \frac{\partial^2 u}{\partial x_k^2} - 2 \sum_{k=1}^{n-1} \frac{\partial^2 u}{\partial x_k \partial x_{k+1}} = 0;$$

$$(4) \frac{\partial^2 u}{\partial x_1^2} - 2 \sum_{k=2}^n (-1)^k \frac{\partial^2 u}{\partial x_{k-1} \partial x_k} = 0.$$